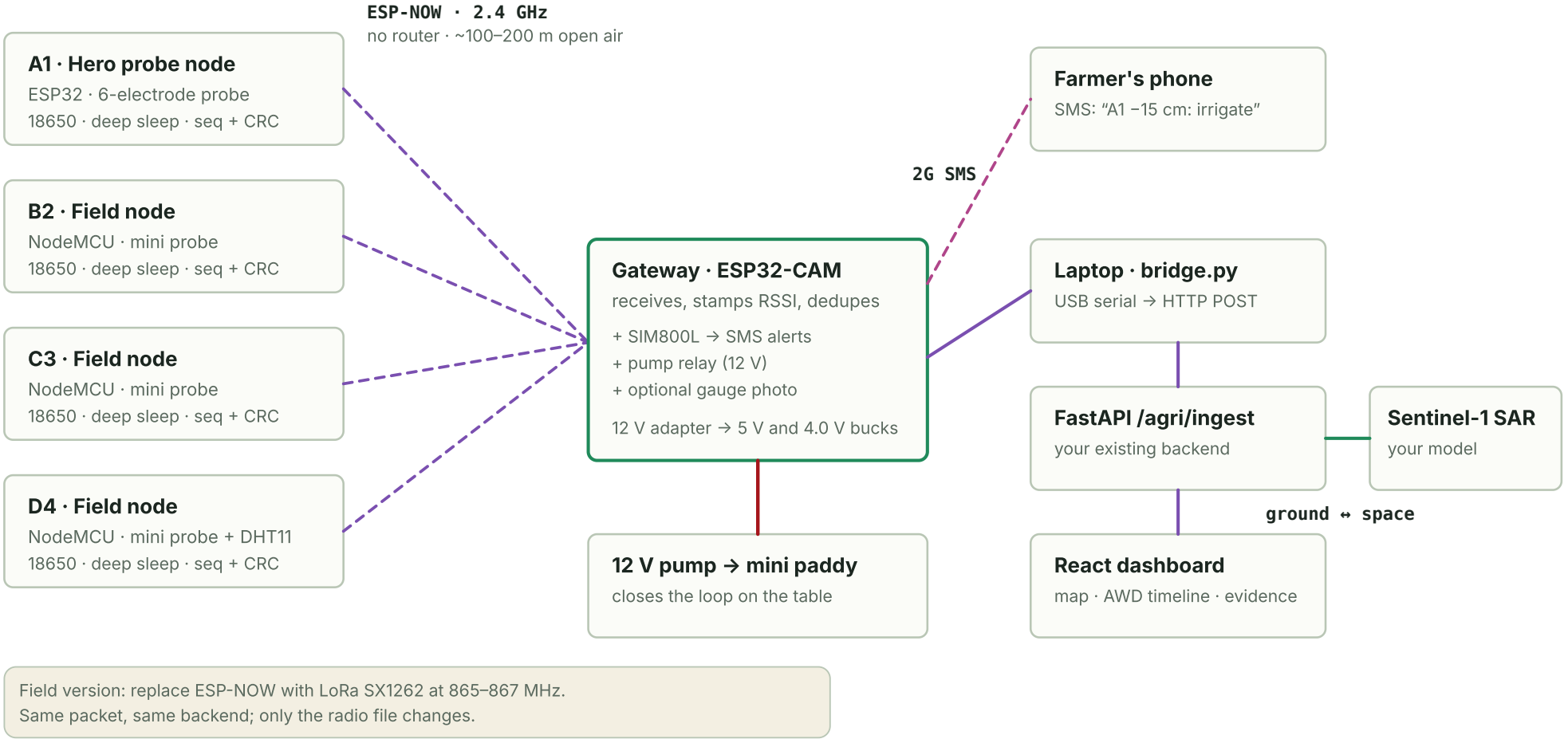
AGRI-GRID hackathon prototype

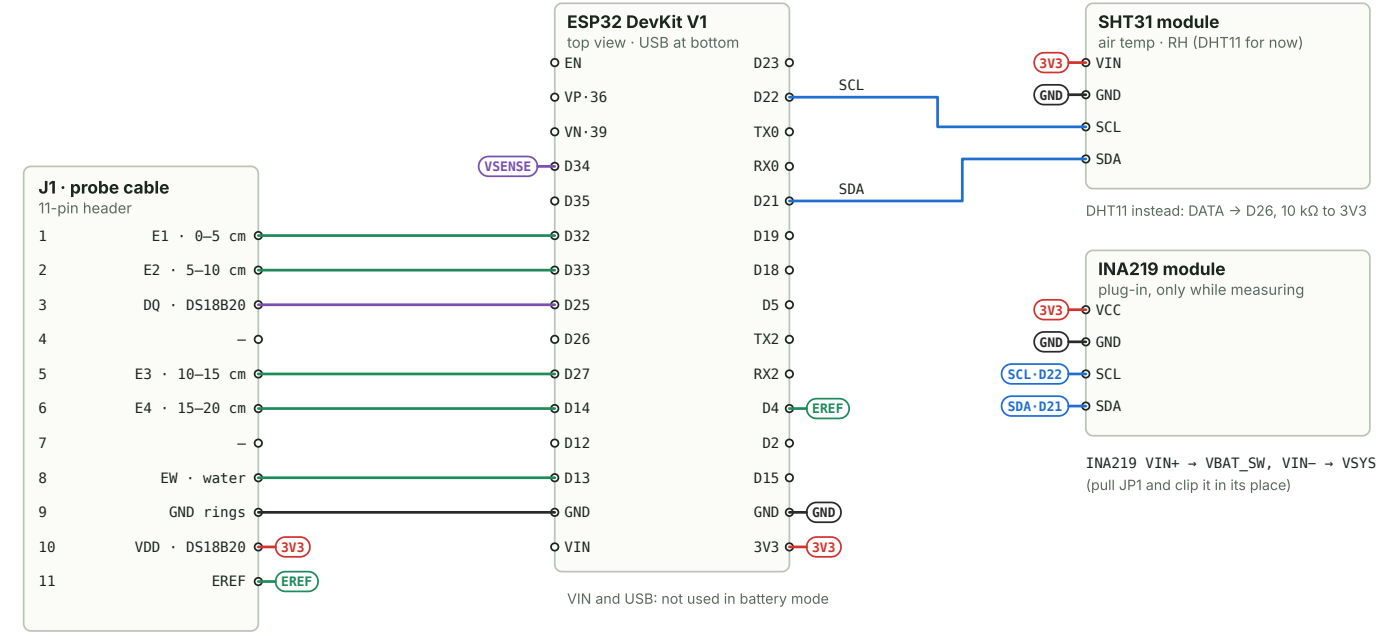
Four ESP nodes with 3D-printed dielectric probes, an ESP-NOW gateway that sends SMS and drives a pump, and the existing FastAPI backend. Everything here uses parts you already own, plus the short list at the end.

AGRI-GRID prototype · how data moves

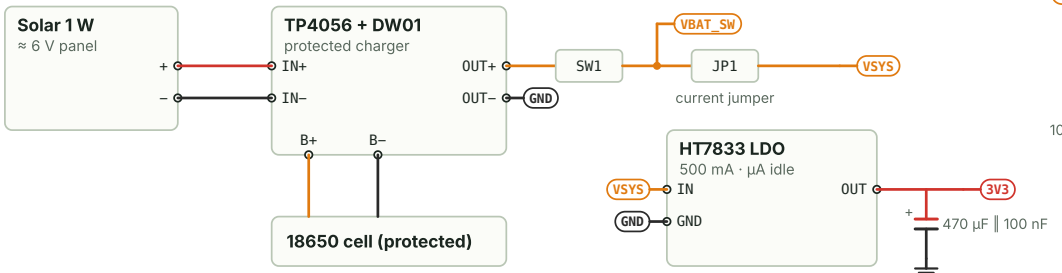


Node A1 · hero probe node (ESP32 DevKit V1, 30-pin)

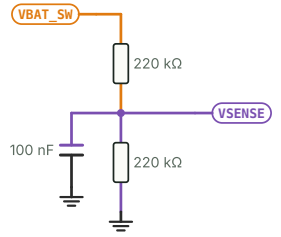
Pills are net names: every pill with the same name is the same wire.



Power path



Battery sense



DS18B20 pull-up



Battery mode feeds the 3V3 pin directly. Switch SW1 off before plugging in USB, every time.
Pull the DevKit's power LED; then measure sleep current across JP1 with a multimeter on 200 µA.

Node A1 pin map (ESP32 DevKit V1, 30-pin)

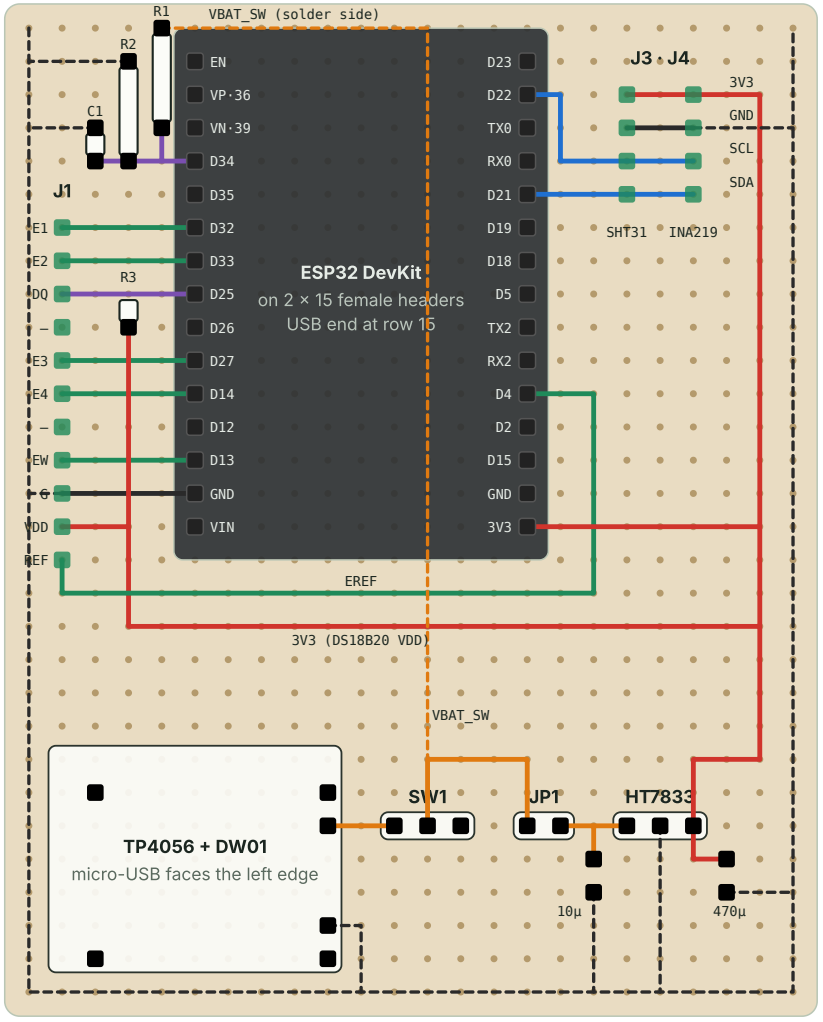
Signal	GPIO	DevKit pin	Note
E1 soil 0–5 cm (SAR-matched)	GPIO32 · T9	D32	touch electrode
E2 soil 5–10 cm	GPIO33 · T8	D33	touch electrode
E3 soil 10–15 cm	GPIO27 · T7	D27	touch electrode
E4 soil 15–20 cm (AWD line)	GPIO14 · T6	D14	touch electrode
EW water strip 0 to +100 mm	GPIO13 · T4	D13	touch electrode
EREF sealed reference (in cap)	GPIO4 · T0	D4	drift / temperature reference
DS18B20 data (soil temp, inside probe)	GPIO25	D25	4.7 kΩ pull-up to 3V3
I²C SDA (SHT31 + INA219)	GPIO21	D21	module pull-ups
I²C SCL (SHT31 + INA219)	GPIO22	D22	module pull-ups
Battery sense (220k / 220k divider)	GPIO34	D34	ADC1, input only
3.3 V in (from HT7833)	3V3	3V3	battery mode: do NOT also plug USB
Ground	GND	GND	

Why these pins

- All six electrodes use ESP32 touch channels that are not boot-strapping pins (GPIO 0, 2, 5, 12, 15 are avoided).
- The ESP32-C3 has **no touch hardware**. Any node that reads the probe this way must be a classic ESP32 or an ESP32-S3.
- Battery sense is on GPIO34, an ADC1 pin, so it still works while the radio is on.
- HT7833 (500 mA) instead of a 250 mA LDO: an ESP32 transmit burst can pull 300 mA or more.

Node A1 · perfboard layout (7 × 9 cm, 24 × 30 holes, viewed from the component side)

Checked by script: no shared holes between nets, no overlapping wires, every net connected.
Solid = insulated jumper on top. Dashed = bare wire on the solder side. Dots = solder joints.



Wire colours

- 3V3
- battery · VBAT_SW · VSYS
- GND
- probe electrodes · EREF
- I²C
- DQ · VSENSE

R1, R2 = 220 kΩ (battery divider) · C1 = 100 nF
R3 = 4.7 kΩ standing (DS18B20 pull-up)
10µ / 470µ = electrolytics, + toward the wire

Flying leads, no holes: solar panel → TP4056 IN±,
18650 holder → TP4056 B±.

Electrode jumpers J1 → ESP32 are short and straight.
Glue them down: a wire that moves changes the
capacitance you are trying to measure.

Header rows 10 holes apart (25.4 mm) suit most
30-pin DevKits; some clones are 9. Measure first.

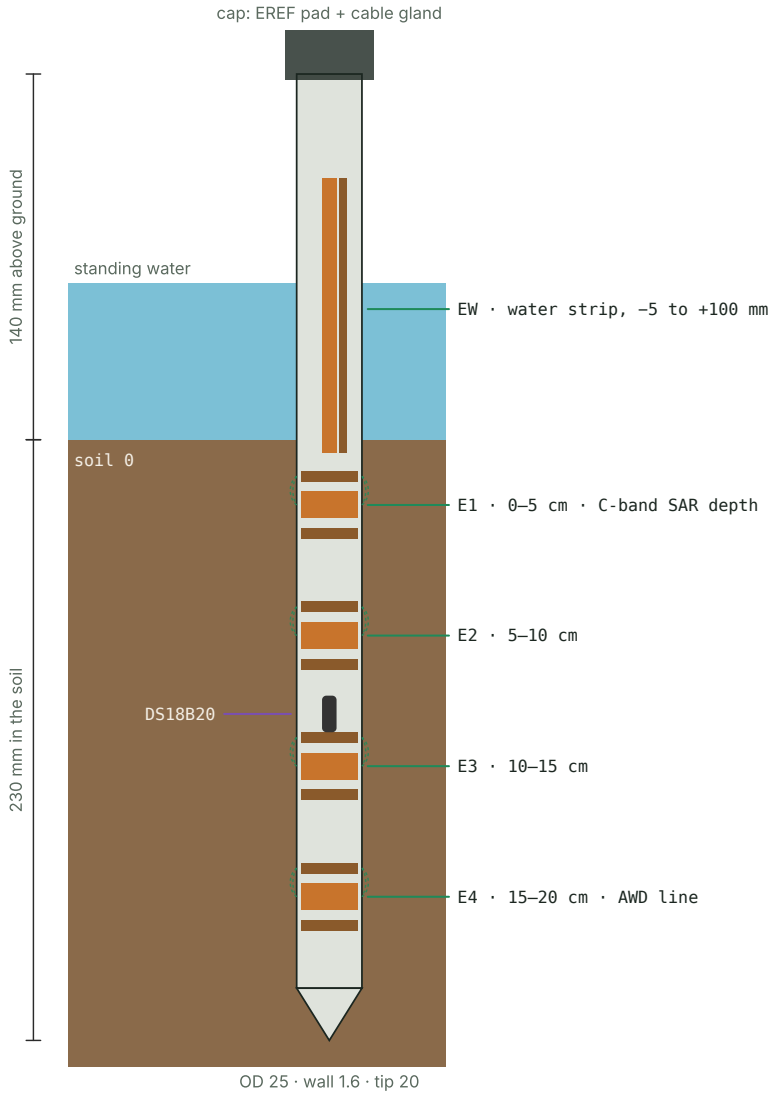
LDO pin order (IN / GND / OUT) differs between
HT7833 packages. Follow your part's datasheet.

TP4056 pad order varies between module versions.
Follow the silkscreen on yours.

Measure sleep current: pull JP1, put the
multimeter (200 µA range) across its two pins.

Dielectric profile probe · PETG, clamshell print

Copper tape on the INSIDE wall. The 1.6 mm PETG wall is the insulator: no metal touches soil or water.



How one electrode reads moisture

Each sense band (10 mm) sits between two GND bands (4 mm).
Its electric field fringes out through the PETG wall into the soil.
Water has a dielectric constant near 80; dry soil about 3–5.
More water → higher capacitance → ESP32 touch count changes.
Topp et al. (1980) turns permittivity ε into water content θ.
Each band reads a thin shell of soil, roughly as far out as
the electrode gap: a few millimetres. Pack soil firmly around it.

Print and build

Print as two half-shells, flat on the bed, no supports.
Lower section 230 mm, upper 140 mm, joined by a sleeve.
Copper tape on the inner face of ONE half-shell; wires up
the other half. Glue halves with epoxy, seal the tip.
Bands at −25, −75, −125, −175 mm (centres).
Black PETG is fine underground. Paint the part above
ground white so the sun does not heat the water strip.
Bed under 235 mm? Split the lower section in two.

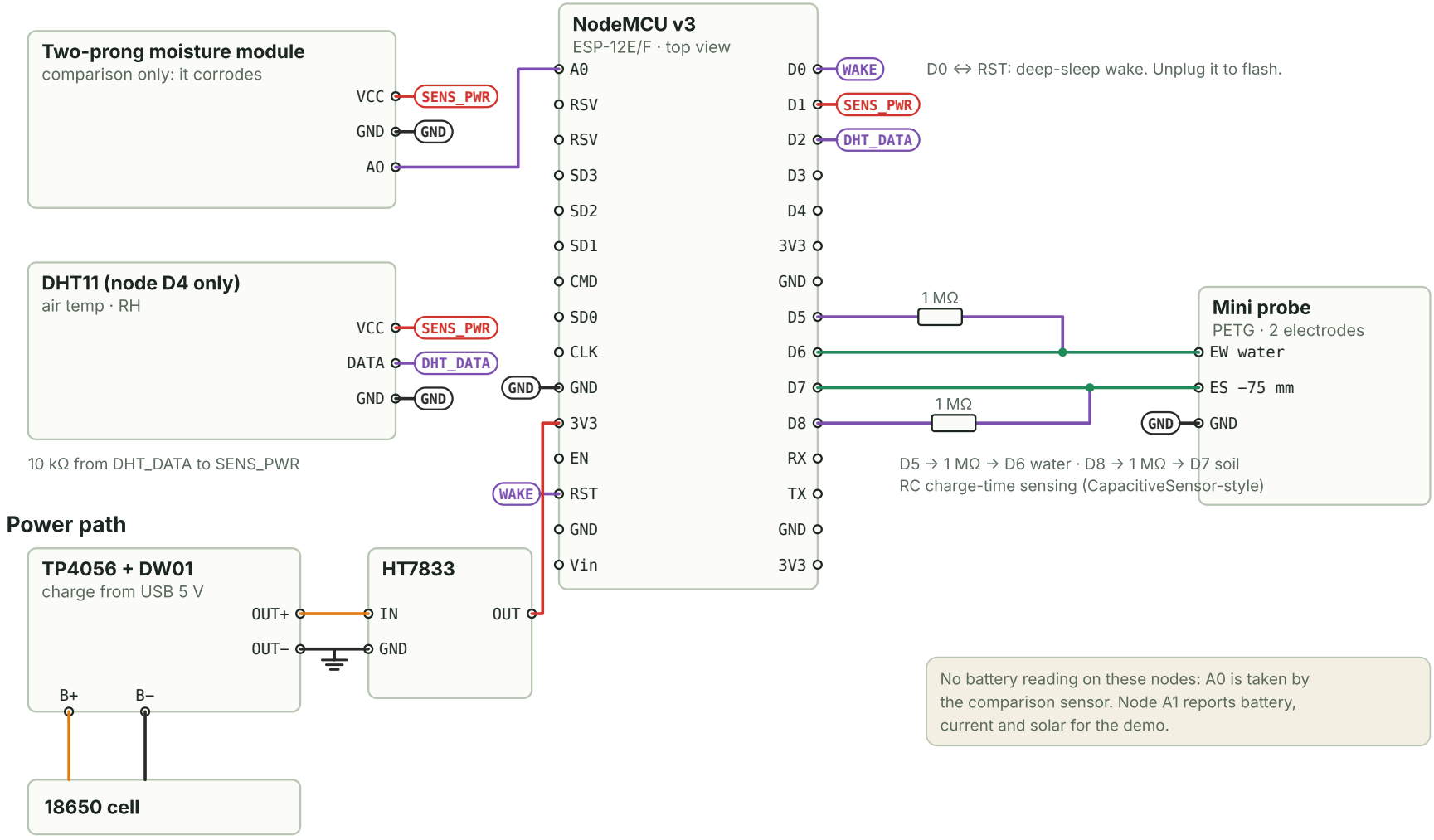
Why it matters for AWD

The water strip reads standing water above the surface.
Bands E1–E4 keep reading after the water disappears,
so the station sees the −15 cm irrigate point that an
ultrasonic sensor above the field cannot.

Field nodes B2–D4 carry a mini version: EW strip + one band (−75 mm), read with RC timing (1 MΩ) on the ESP8266.

Nodes B2 / C3 / D4 · field node (NodeMCU ESP8266)

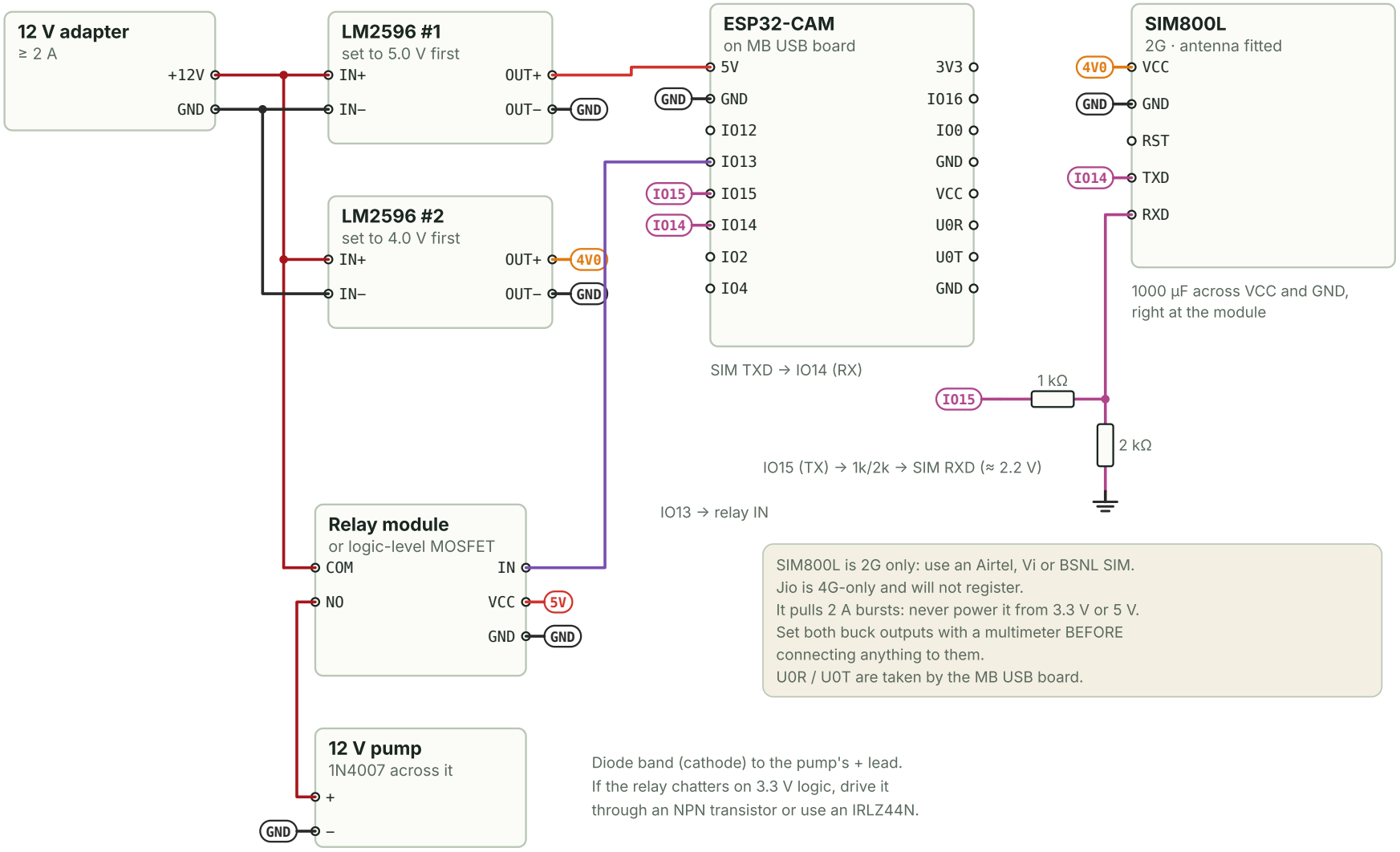
Battery-only, deep sleep, ESP-NOW to the gateway. Node D4 also carries the DHT11.



Feed the 3V3 pin, not Vin: Vin goes through the board's AMS1117, which drops too much from one cell.

Gateway · ESP32-CAM + SIM800L SMS + 12 V pump

Runs from your 12 V adapter. The MB USB board links it to the laptop for the data bridge.



Field node pin map (NodeMCU v3)

Signal	GPIO	NodeMCU pin	Note
Mini probe EW (water strip)	GPIO12	D6	receive; 1 MΩ from D5
Mini probe ES (soil −75 mm)	GPIO13	D7	receive; 1 MΩ from D8
RC send, water	GPIO14	D5	drive high, time D6
RC send, soil	GPIO15	D8	board pull-down keeps boot safe
Sensor power (moisture module, DHT11)	GPIO5	D1	HIGH only while reading
DHT11 data (node D4)	GPIO4	D2	10 kΩ to SENS_PWR
Two-prong module AO	ADC	A0	0–3.3 V on NodeMCU
Deep-sleep wake	GPIO16	D0 → RST	remove to flash
3.3 V in from HT7833	3V3	3V3	not Vin

Gateway pin map (ESP32-CAM)

Signal	GPIO	ESP32-CAM pin	Note
SIM800L TXD → ESP RX	GPIO14	IO14	UART2 RX
ESP TX → divider → SIM800L RXD	GPIO15	IO15	1 kΩ / 2 kΩ
Pump relay / MOSFET	GPIO13	IO13	HIGH = pump on
5 V in	5V	5V	from LM2596 #1
USB serial to laptop	U0R/U0T	MB board	bridge.py reads this

Bring-up order

1. **Set the bucks first.** Power each LM2596 from 12 V with nothing attached. Turn the trimmer until the multimeter reads 5.00 V and 4.00 V. Only then wire them up.
2. **ESP32 alone, on USB.** Flash a sketch that prints touchRead() for GPIO 32, 33, 27, 14, 13 and 4 once a second. Touch each J1 pin with a finger: the count should jump.
3. **Probe on the bench.** Plug in the probe. Record counts in air, in dry soil and in a glass of water. Water must give the biggest change. If a band does not move, check its wire.
4. **Battery path.** Charge the 18650 through the TP4056 (red LED → blue). Check OUT+ is 3.7–4.2 V and the HT7833 gives 3.30 V with nothing attached.
5. **Battery mode.** Unplug USB. Switch SW1 on with the ESP32 fitted. The board should boot from the battery. Never have USB and SW1 on together.
6. **Measure sleep current.** Pull JP1. Put the multimeter (200 µA or 2 mA range) across its pins and let the node sleep. Write the number down: it is your first real power measurement.
7. **Field nodes.** Flash each NodeMCU with D0–RST unplugged, then fit the wire. Confirm the gateway prints a packet from each node with rising seq numbers.
8. **Gateway + SMS.** Insert an Airtel, Vi or BSNL SIM. Send AT, then AT+CREG? until it reports registered. Send one test SMS to your phone before the demo day.

Calibrating the probe

1. Take 5–6 soil samples from the demo field, from bone-dry to saturated (add water in steps).
2. Pack each sample around the probe in a cup of known volume V. Record the touch counts for the band.
3. Weigh the wet sample (m_wet). Dry it at 105 °C for 24 h, or in a microwave in 1-minute bursts until the weight stops changing. Weigh again (m_dry).
4. Volumetric water content: $\theta = (m_{wet} - m_{dry}) / (\rho_{water} \cdot V)$, with $\rho_{water} = 1\ g/cm^3$.
5. Plot θ against the counts and fit a straight line or a quadratic. That curve is your calibration, and the slide judges remember.
6. Water strip: stand the probe in a bucket and pour in 0, 20, 40, 60, 80 and 100 mm. Fit counts against depth.
7. Repeat one point at a different temperature to see the drift; the DS18B20 lets you correct for it.

Already on your bench

ESP32 DevKit, ESP32-CAM, 3 × NodeMCU, 3 × ESP-12F, INA219, DHT11, two-prong moisture module, relay module, 12 V pump, 12 V adapter, 4–6 × 18650 with holders, 1 W panel, perfboard, resistors, LEDs, solder station, black PETG and the printer. SIM800L from your friend.

Short list to buy (rough prices; check before ordering)

Part	Qty	Price	Why
HT7833 (or AP2112K-3.3) LDO	4	≈ ₹20 each	one per battery node
TP4056 + DW01 protected module	4	≈ ₹30–50 each	one per battery node
LM2596 buck module	2	≈ ₹80–120 each	gateway 5.0 V and 4.0 V rails
Waterproof DS18B20	2	≈ ₹120 each	soil temperature, drift correction
SHT31 module	1	≈ ₹300	replaces DHT11 outdoors
Copper foil tape, 10 mm	1 roll	≈ ₹150	probe electrodes
Clear epoxy	1	≈ ₹100	seal the probe halves and tip
1000 µF / 470 µF / 10 µF electrolytics	few	≈ ₹50	SIM800L and LDO bulk caps
Female headers, 11-pin and 4-pin headers, slide switch	set	≈ ₹80	hero perfboard
Later: RFM95W-868 or Ra-01H pair + 868 MHz antennas	2	≈ ₹1,400–1,600	one real LoRa link with RSSI

Prices are typical Indian online listings, not quotes. LoRa can wait: ESP-NOW runs the demo, and the firmware keeps the radio behind one interface so swapping later changes one file.